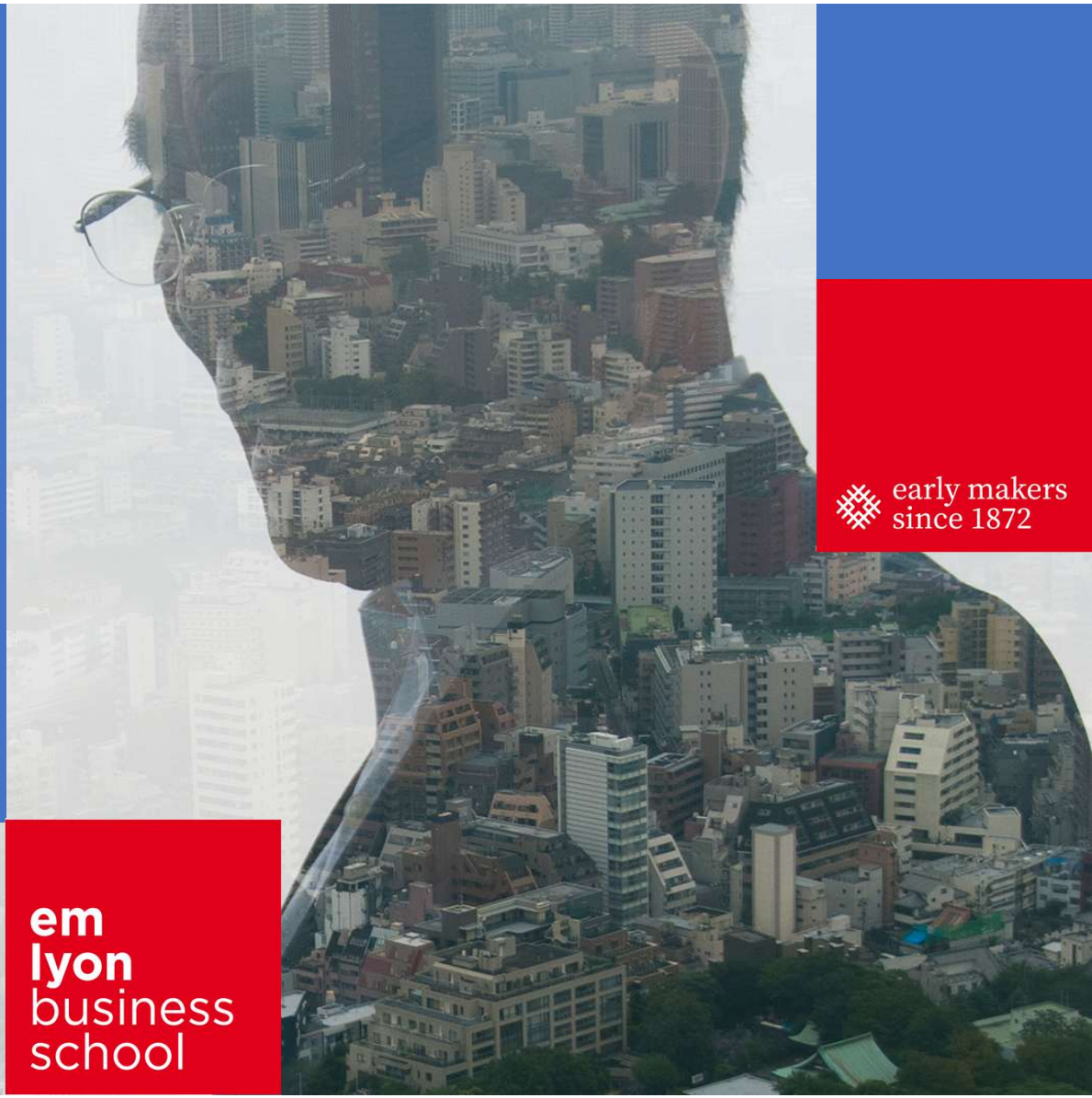
Introduction to Final Project (50%)



early makers since 1872

em lyon business school

Professors (Visiting):

Dr. Hareesh (何玄)

Zhong WeiBing (仲伟冰)



Requirements

R for Data Science — Final Project Requirements

The final project is a **group-based data science project** worth **50% of the course grade**. It requires you to apply the full R data science workflow: identify an industry context, build a scientific background from academic literature, obtain a relevant dataset, conduct exploratory data analysis in R, design a dashboard informed by both the literature and the EDA, implement the dashboard as a Shiny application, and document your process and learning in a referenced report.
The project should demonstrate the course learning objectives: tidy and reshape data, summarize and visualize data, integrate external data, build interactive Shiny applications, and communicate insights reproducibly.

Project Structure — Required Steps

Step 1: Group Formation and Industry Selection

- Form a group as assigned by the instructor.
- Select an **industry or sector** (e.g., retail, finance, healthcare, education, transportation, energy, public policy, sports, hospitality, or another approved field).
- Define the **business or policy context** clearly: Who is the audience? What decisions could the data inform?
- Submit a short **industry selection note** to the instructor for approval.

Deliverable: Industry selection and problem context (1 page).

Project Structure — Required Steps

Step 2: Scientific Background and Literature Review

Step 3: Dataset Identification and Acquisition

Step 4: Exploratory Data Analysis (EDA) in R

Step 5: Thought Framework and Dashboard Design

Step 6: Dashboard Execution as a Team

Step 7: Final Report with References

Step 8: Final Presentation

Session	Milestone	Due / Activity
Session 5	Project overview, academic reading, literature review methodology	Submit **project design plan and timeline**; industry selection approved
Session 6	Data storytelling, project design, code design	Submit **literature review summary and dataset description**; dashboard design specification draft
Session 7	Project development, output review, scientific rigor	Submit **EDA notebook and dashboard prototype** for code/output review
Session 8	Supervised teamwork, final presentation and submission	Submit **final report, Shiny app, presentation, and individual reflections**; present in class

recommend:

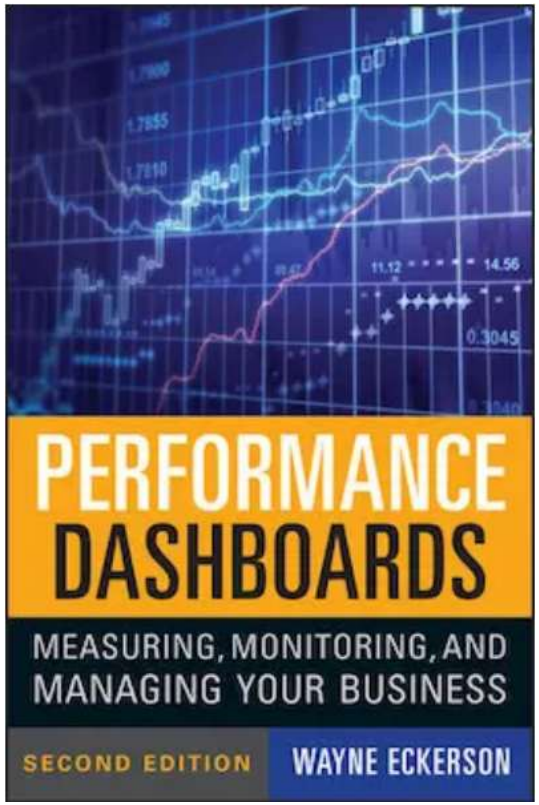
Stop seeing datasets as spreadsheets and start seeing:
Customers, workers, transactions, delays, decisions, operational constraints, incentives

> “What kind of real-world activity generated this row of data?”



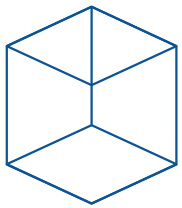
Dashboards for Business Informatics
商业信息学仪表板

Books



《Performance Dashboards: Measuring, Monitoring, and Managing Your Business》

Author: Wayne W. Eckerson
Publication Year：2010



Dashboard

Definition：

- Dashboard can translates the organization's strategy into objectives, metrics, initiatives, and tasks customized to each group and individual in the organization.
  仪表盘可以将组织战略转化为目标、指标、计划和任务，并为组织中的每个小组和个人量身定制。

Functions：

- It provides timely information and insights that enable business users to improve decisions, optimize processes and plans, and work proactively.
  它提供及时的信息和见解，使业务用户能够改进决策、优化流程和计划，并积极主动地开展工作。
- It communicates strategic objectives and enables businesspeople to measure, monitor, and manage the key activities and processes needed to achieve their goals.
  它传达战略目标，使业务人员能够衡量、监控和管理实现目标所需的关键活动和流程。

Resource：《Performance Dashboards: Measuring, Monitoring, and Managing Your Business》P4

Three Types of Dashboard

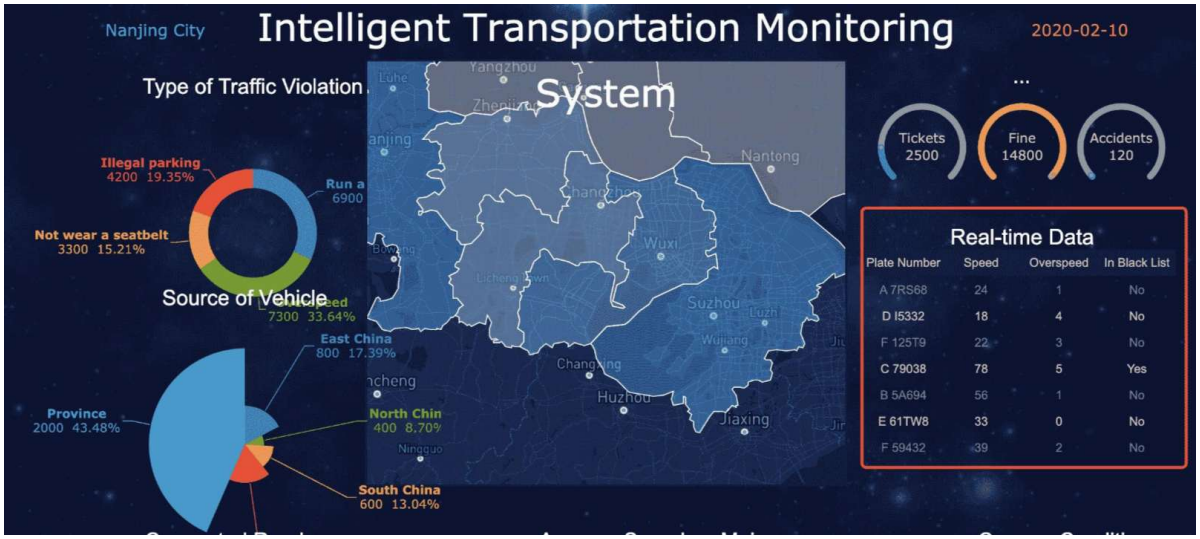
Operational Dashboard

- Operational dashboards are used to control operational activity and make sure processes stay within prescribed limits for productivity, quality, and efficiency.

操作仪表板用于控制操作活动，确保流程在规定范围内提高生产率、质量和效率。

- Operational dashboards emphasize **monitoring** more than analysis or management.

运行仪表盘强调的是监控，而不是分析或管理。



Resource: 《Performance Dashboards: Measuring, Monitoring, and Managing Your Business》P17

Three Types of Dashboard

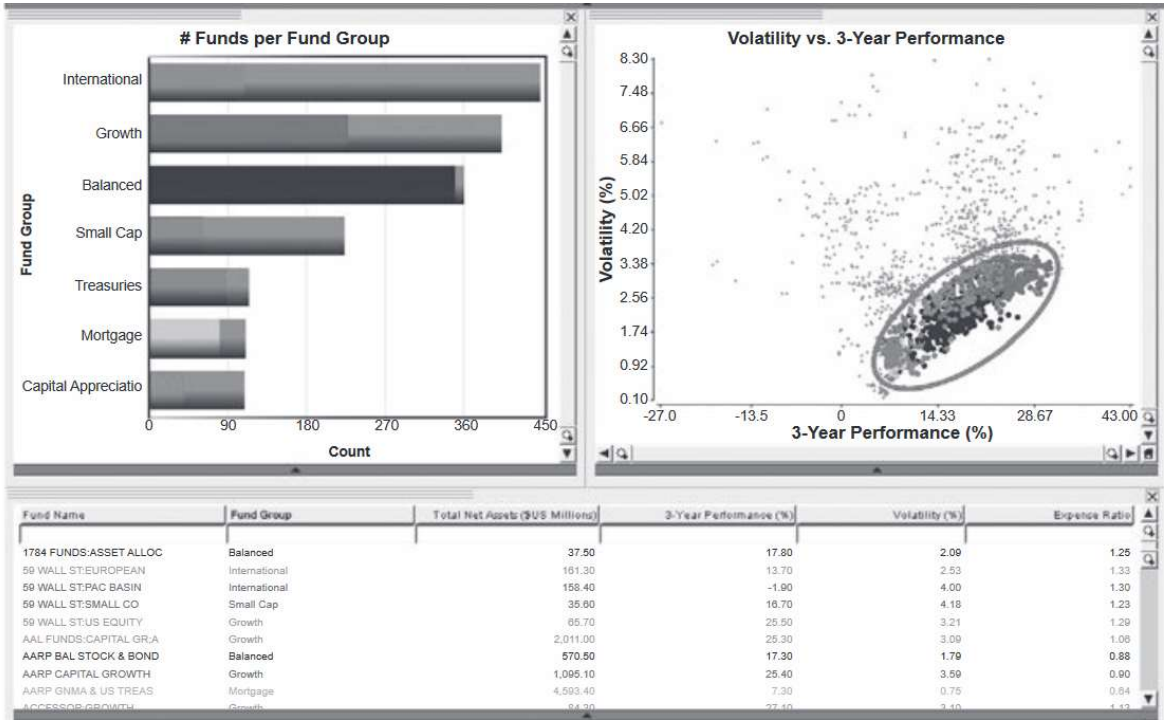
Tactical Dashboard

- Tactical dashboard are used to monitor and manage departmental processes and projects.

  战术仪表板用于监控和管理部门流程和项目。

- Tactical dashboard tend to emphasis **analysis** more than monitoring or management.

  战术仪表板往往更强调分析，而不是监测或管理。



Fund Name	Fund Group	Total Net Assets ($US Millions)	3-Year Performance (%)	Volatility (%)	Expense Ratio
1784 FUNDS:ASSET ALLOC	Balanced	37.50	17.80	2.09	1.25
59 WALL ST:EUROPEAN	International	161.30	13.70	2.53	1.33
59 WALL ST:PAC BASIN	International	158.40	-1.90	4.00	1.30
59 WALL ST:SMALL CO	Small Cap	35.60	16.70	4.18	1.23
59 WALL ST:US EQUITY	Growth	65.70	25.50	3.21	1.29
AAL FUNDS:CAPITAL GR:A	Growth	2,011.00	25.30	3.09	1.06
AARP BAL STOCK & BOND	Balanced	570.50	17.30	1.79	0.88
AARP CAPITAL GROWTH	Growth	1,095.10	25.40	3.59	0.90
AARP GNMA & US TREAS	Mortgage	4,593.40	7.30	0.75	0.84
ACCESSOR GROWTH	Growth	84.20	27.40	3.40	1.12

Resource：《Performance Dashboards: Measuring, Monitoring, and Managing Your Business》P17、P114

Three Types of Dashboard

Strategic Dashboard

- Strategic dashboard are used to monitor the execution of strategic objectives

  战略仪表板用于监测战略目标的执行情况

- Strategic dashboard tend to emphasis **management** more than monitoring or analysis.

  运行仪表盘强调的是监控，而不是分析或管理。

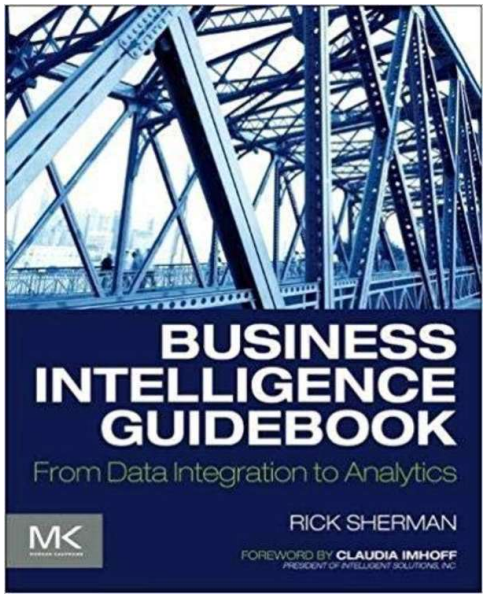


Resource: 《Performance Dashboards: Measuring, Monitoring, and Managing Your Business》P17



Introduction to Business Informatics (BI)

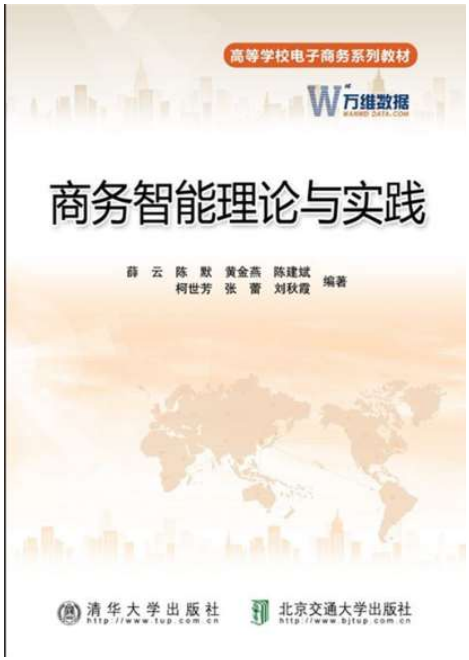
Books



《Business Intelligence Guidebook: From Data Integration to Analytics》

Author: Rick Sherman

Publication Year：2014



《商务智能理论与实践》

Author: 薛云 等

Publication Year：2019

Background

- Organizations collect a lot of data every day, both internally and externally, and problems arise when there is more data than the organization can handle.

企业每天都会从内部和外部收集大量数据，当数据多到企业无法处理时，就会出现问题。

- Businesses may be awash with data, but that does not necessarily mean they have useful information.

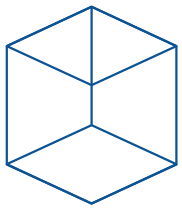
企业可能拥有大量数据，但这并不一定意味着它们拥有有用的信息

Businesses need a useful tool to help them get valuable information from large amounts of data.

企业需要一个有用的工具来帮助企业从大量数据中获得有价值的信息。



Resource:《business intelligence guidebook from data integration to analytics compress》P4



Business Informatics (BI)

Definition：

- **Microsoft :** BI is any attempt to capture and analyze corporate data to gain a clearer understanding of the market and customers, to improve corporate processes, and to compete more effectively in the business.

  BI是任何尝试获取、分析企业数据以更清楚地了解市场和顾客，改进企业流程，更有效地参与企业竞争的努力。

- **IBM:** BI is the use of your own data resources to make better business decisions. It includes data access, data and business analysis, and the discovery of new business opportunities.

  BI是指利用已有的数据资源做出更好的商业决策。它包括数据访问、数据和业务分析及发现新的商业机会。

Business Informatics is essentially timely, accurate, high-value, and actionable business insights, and the work processes and technologies used to obtain them.

商业智能的本质是及时、准确、高价值和可操作的业务洞察力，以及用于获取这些洞察力的工作流程和技术。

BI Main Functions

Data Integration 数据集成

Data is the basis for decision-making and analysis, and in order to make the right business decisions, it is necessary to collect these scattered data to form a systematic whole.

数据是决策分析的基础，为了做出正确的经营决策，就需要把这些零散的数据收集起来，形成一个系统的整体。

Operational Analysis 运营分析

Operation analysis refers to the analysis of various types of indicators of the enterprise, timely grasp of the actual situation of the enterprise in all aspects, adjust and reduce the operation of the enterprise.

运营分析指对企业各类指标进行分析，及时掌握企业在各方面的实际情况，调整企业运营情况。

Presentation of information 信息呈现

Information presentation refers to the presentation of collected data in the form of reports, users can understand the current status of the enterprise and market, which is the initial function of BI.

信息呈现是指把收集的数据以报表的形式呈现出来，让用户了解到企业、市场现状，这是 BI 的初步功能。

Strategic Decision Support 战略决策支持

Refers to the development of a rational portfolio strategy based on the operating performance and positioning of each of the company's strategic business units (SBUs).

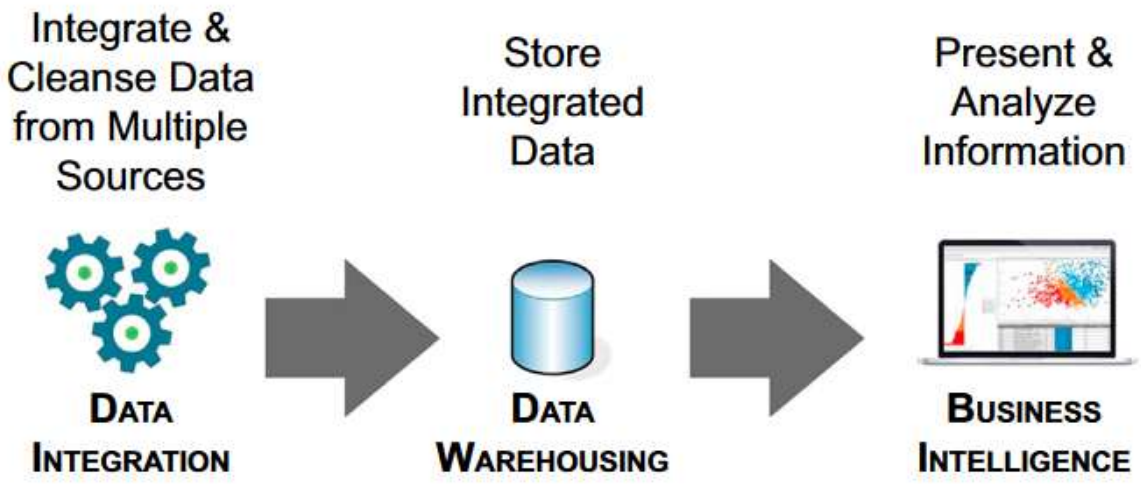
指根据公司各战略业务单元(strategic business unit，SBU)的经营业绩和定位，帮助在制定一种合理的投资组合战略。

Resource:《商务智能理论与实践》P24

Core Building Blocks

Data integration— combining data from different sources and bringing it together to ultimately provide a unified view.

数据整合--将不同来源的数据整合在一起，最终提供统一的视图。




Text Data


ERP/CRM Data


Sensor Data


IoT Data

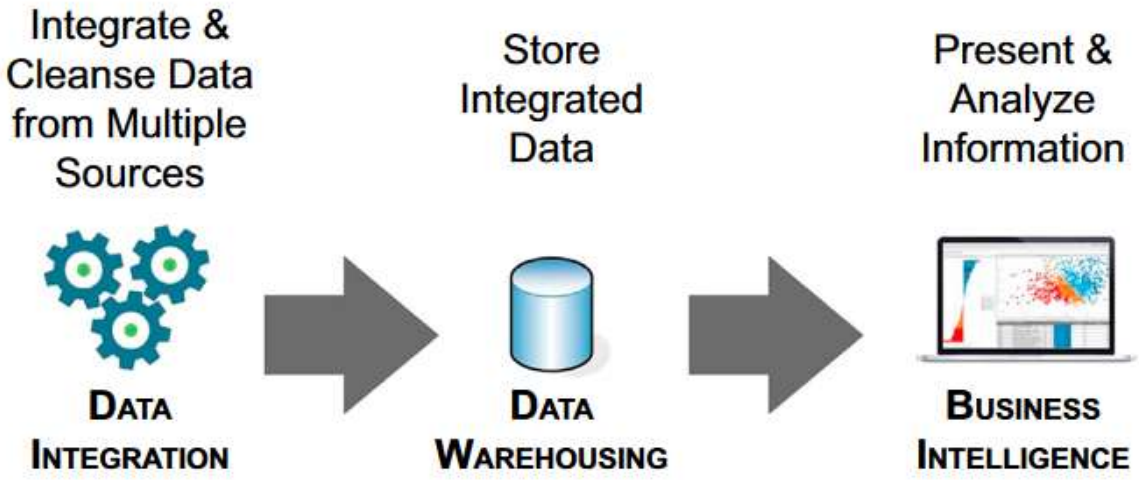
Social Media Data

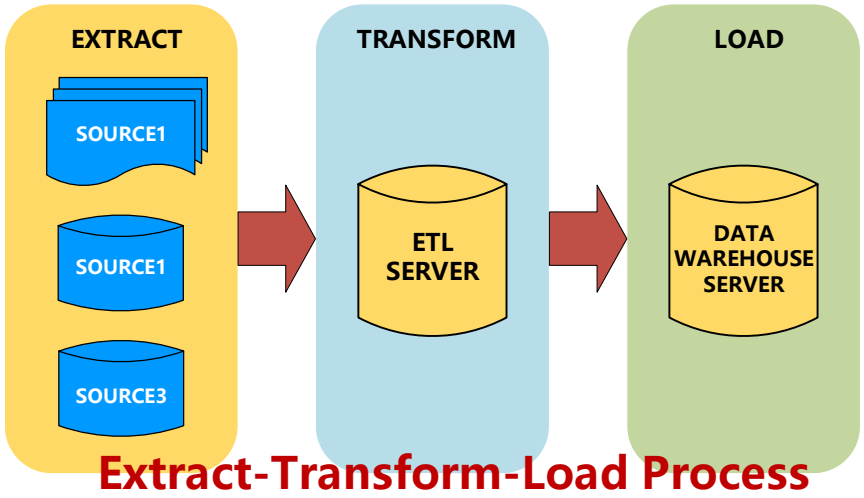
Resource:《business intelligence guidebook from data integration to analytics compress》P4

Core Building Blocks

Data warehousing— the process of storing and staging information, separate from an enterprise's day-to-day transaction processing operations, and optimizing it for access and analysis in an enterprise.

数据仓库--存储和暂存信息的过程，与企业的日常交易处理业务分开，并对其进行优化，以便企业进行访问和分析。





Extract-Transform-Load Process

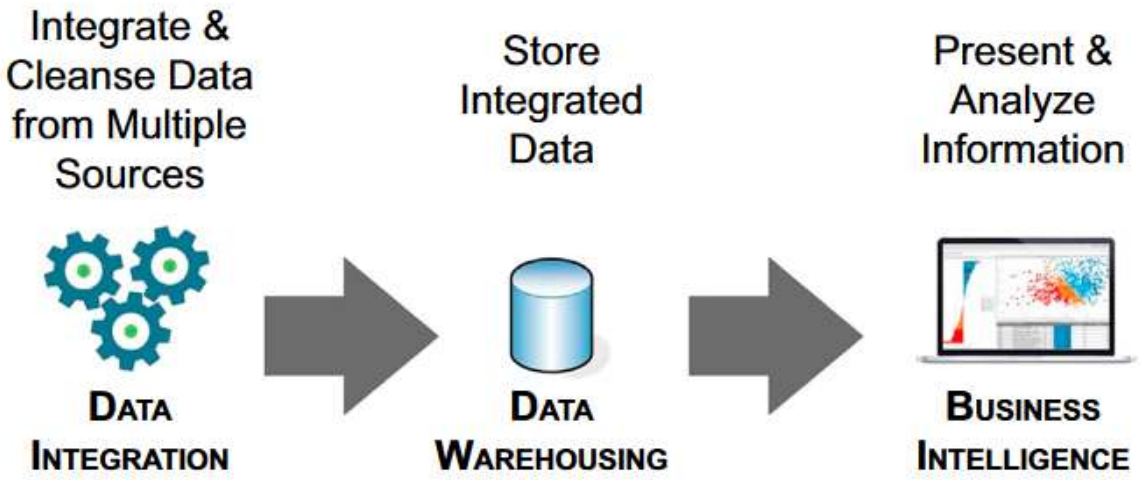
The purpose of ETL is to integrate the scattered, fragmented and inconsistent data in the enterprise to provide an analytical basis for enterprise decision-making

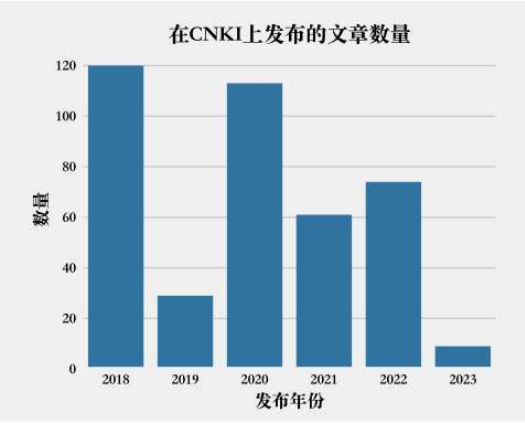
ETL的目的是将企业中分散、零乱、标准不统一的数据整合到一起，为企业的决策提供分析依据

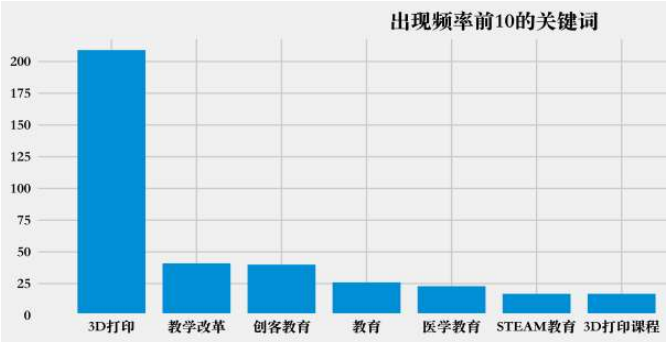
Core Building Blocks

Business Intelligence— to present data to business people so they can use it to gain knowledge. BI enables access and delivery of information to business users. It is the visible portion of the corporate data systems.

商业智能--向业务人员展示数据，以便他们利用数据获取知识。商业智能使企业用户能够访问和传递信息。它是企业数据系统的可见部分。



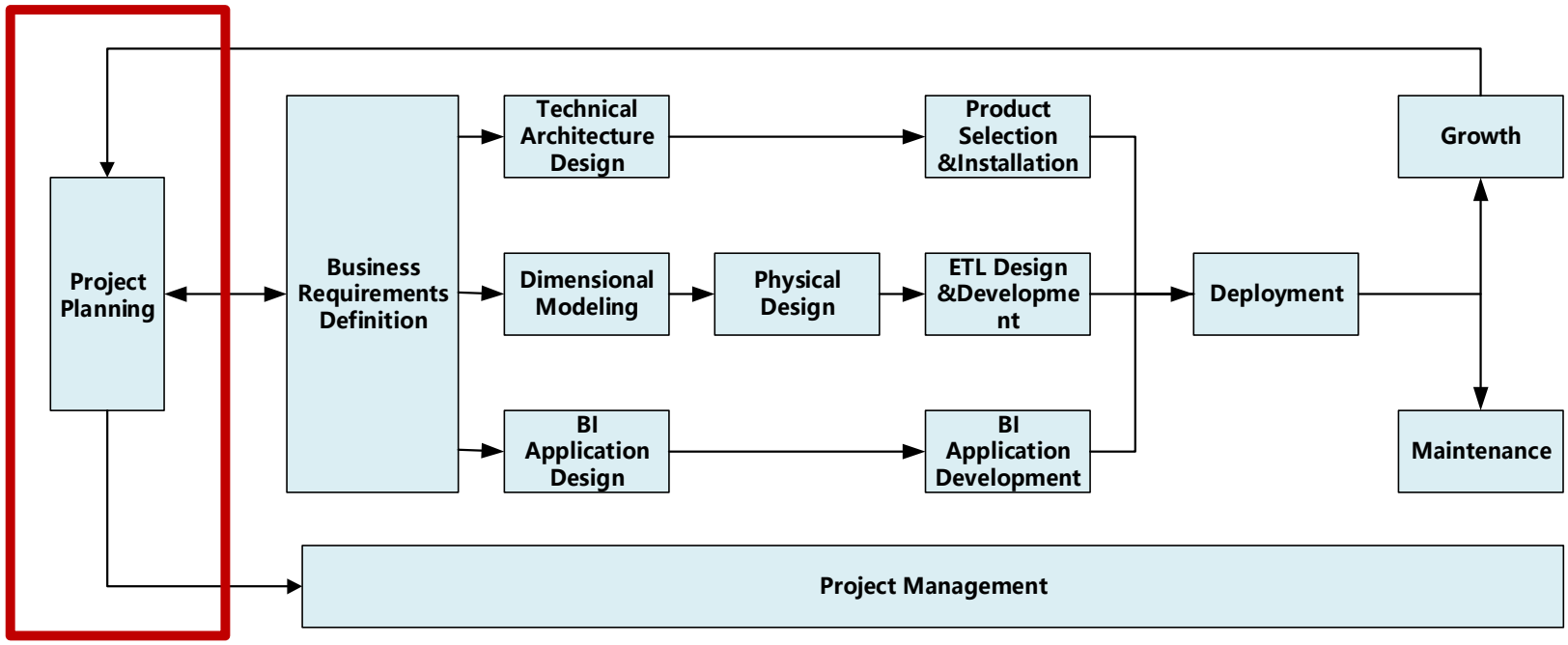






Resource:《business intelligence guidebook from data integration to analytics compress》P4

BI Implementation Process



ETL: Extract, Transform, Load

Project Planning Phase

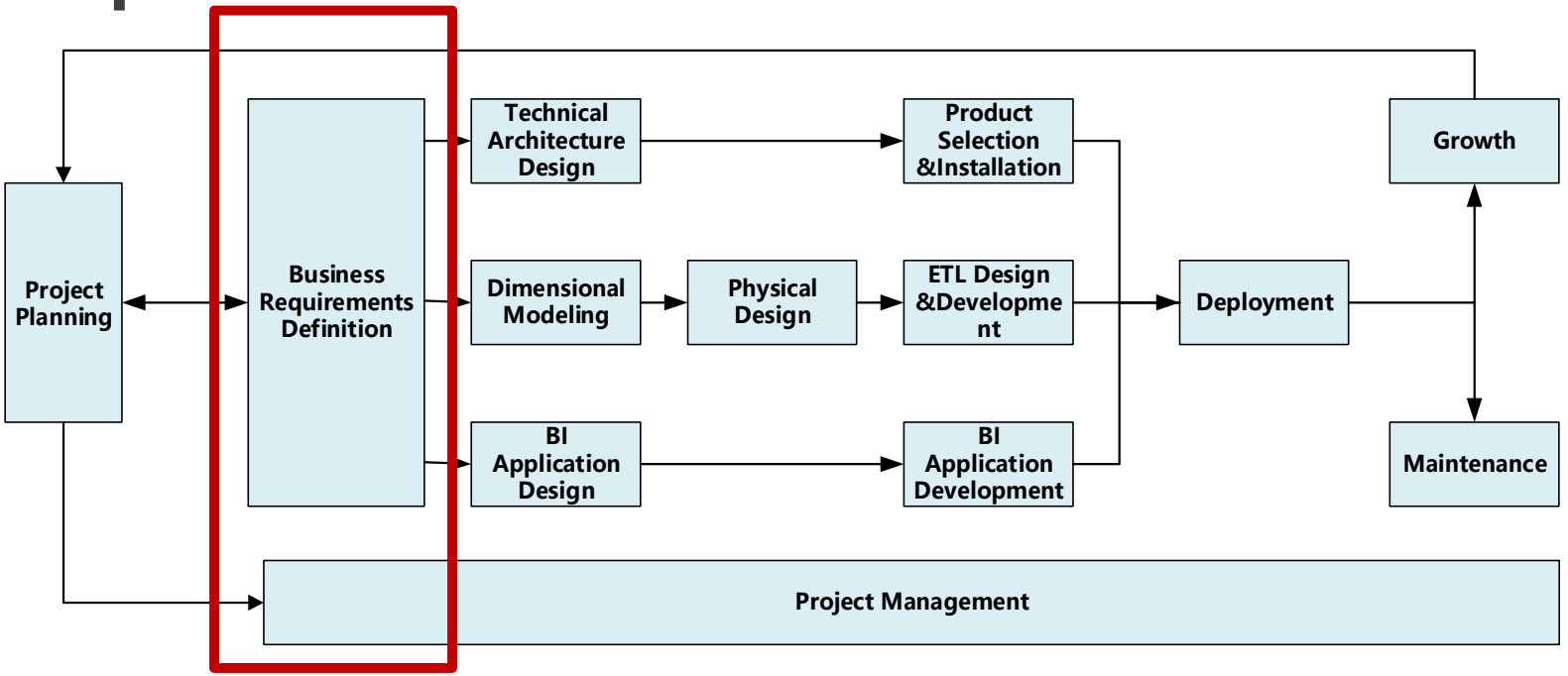
Evaluate whether the enterprise itself has the conditions to implement BI, determine the size and scope of the system, and plan various resources.

评估企业本身是否具备实施BI的条件，确定系统的规模和范围，规划各种资源。

Resource:《商务智能理论与实践》P33

Resource:《商务智能理论与实践》P33

BI Implementation Process



Enterprise Requirements Definition Phase

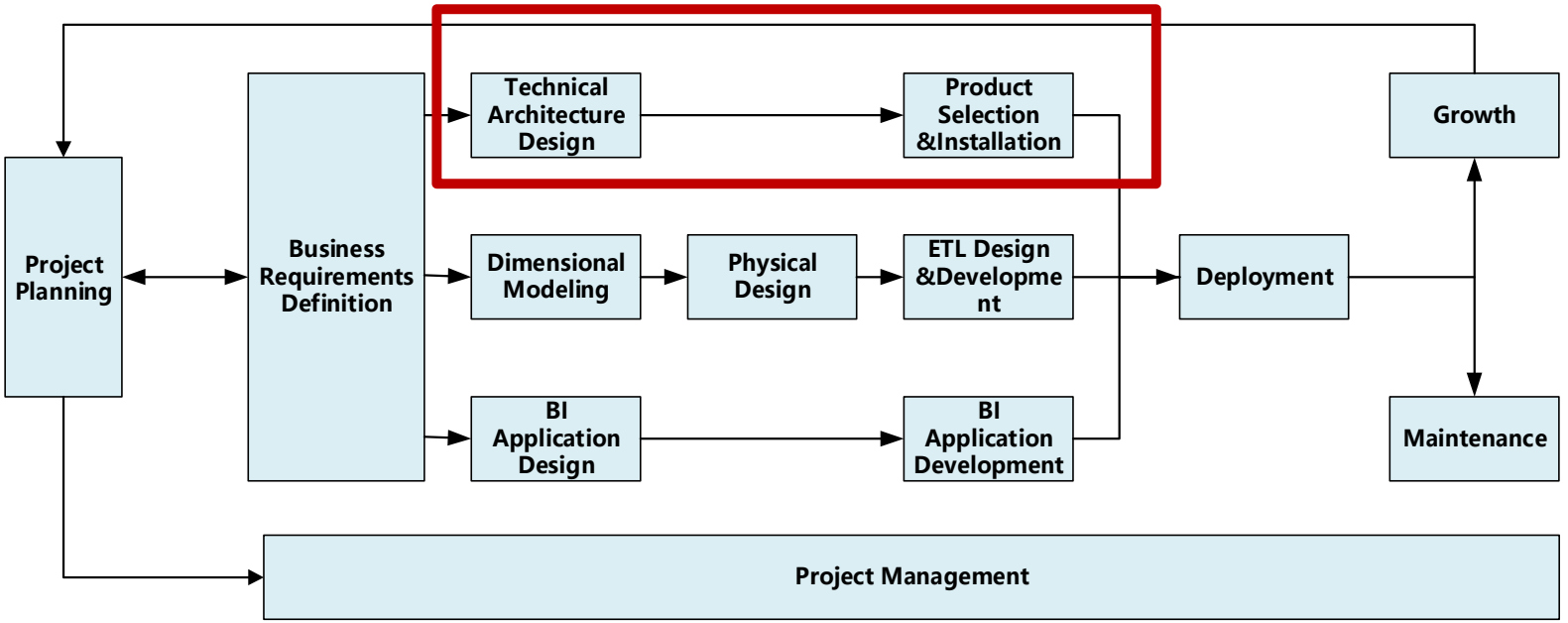
ETL: Extract, Transform, Load

The designer of a BI system should understand the needs of the business and translate those needs into system requirements. The key factor for the success of a BI project is its ability to support business decision-making in terms of its actual business process.

BI系统的设计者应该了解企业的需求并将这些需求转化为系统需求。一个BI项目成功的关键因素是在其实际的商业过程上，是否能够为商务决策提供支持。

Resource:《商务智能理论与实践》P33

BI Implementation Process



Technical Framework Design

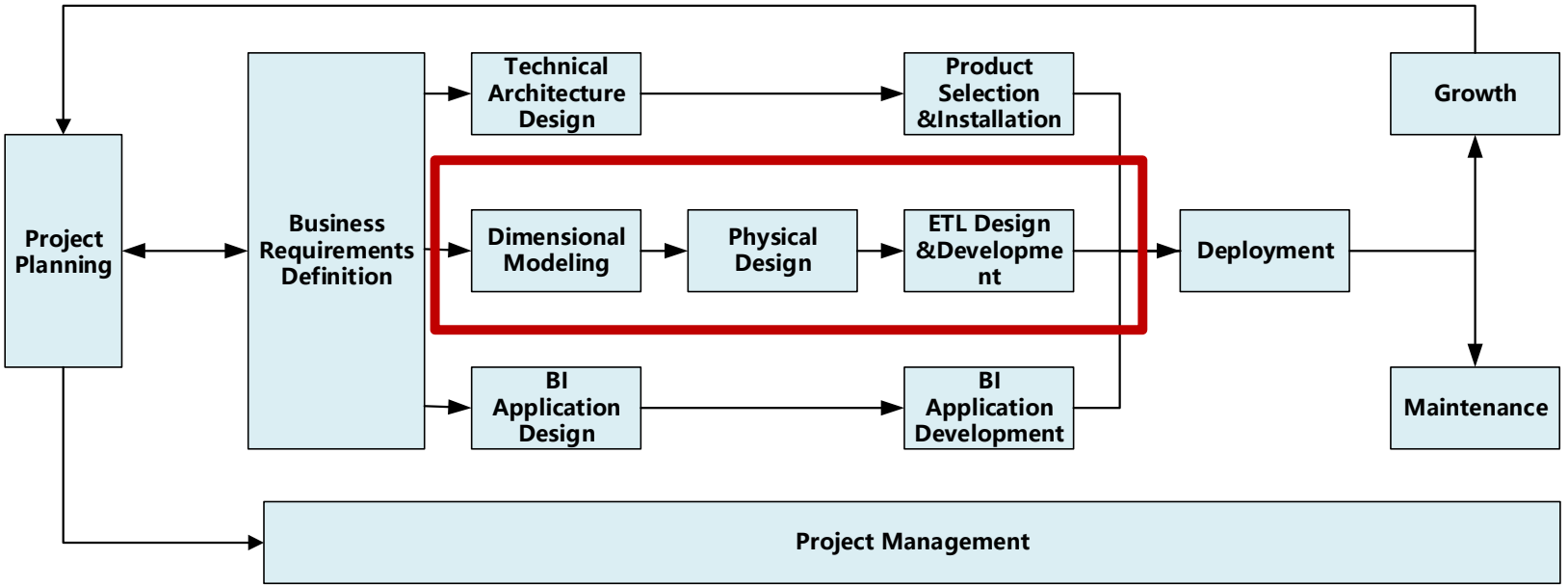
ETL: Extract, Transform, Load

The technology framework design will establish a technology framework to integrate various related technologies; list a series of BI-related products, through the criteria of one, evaluate these products and make the final choice.

技术框架设计将建立一个技术框架，将各种相关技术进行整合;列出一系列的BI相关产品，通过一的标准，对这些产品进行评估，做出最后的选择。

Resource:《商务智能理论与实践》P33

BI Implementation Process



Data Design

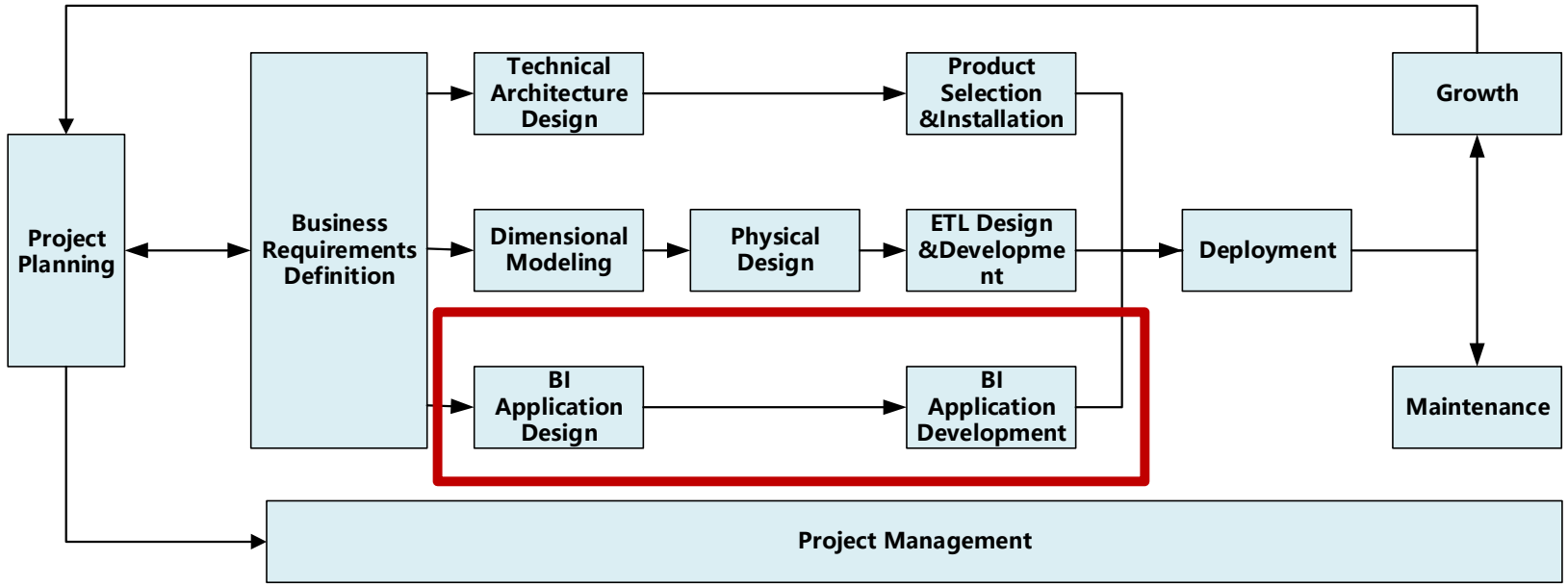
ETL: Extract, Transform, Load

Data design is to transform the enterprise requirements into a multi-dimensional model then design a physical model based on the multi-dimensional model, followed by data extraction, transformation, loading design, and building the actual data warehouse

数据设计是先将企业需求转化成多维模型再根据多维模型设计物理模型，然后是数据抽取、转换、加载设计，建立实际的数据仓库

Resource:《商务智能理论与实践》P33

BI Implementation Process



BI Application Design

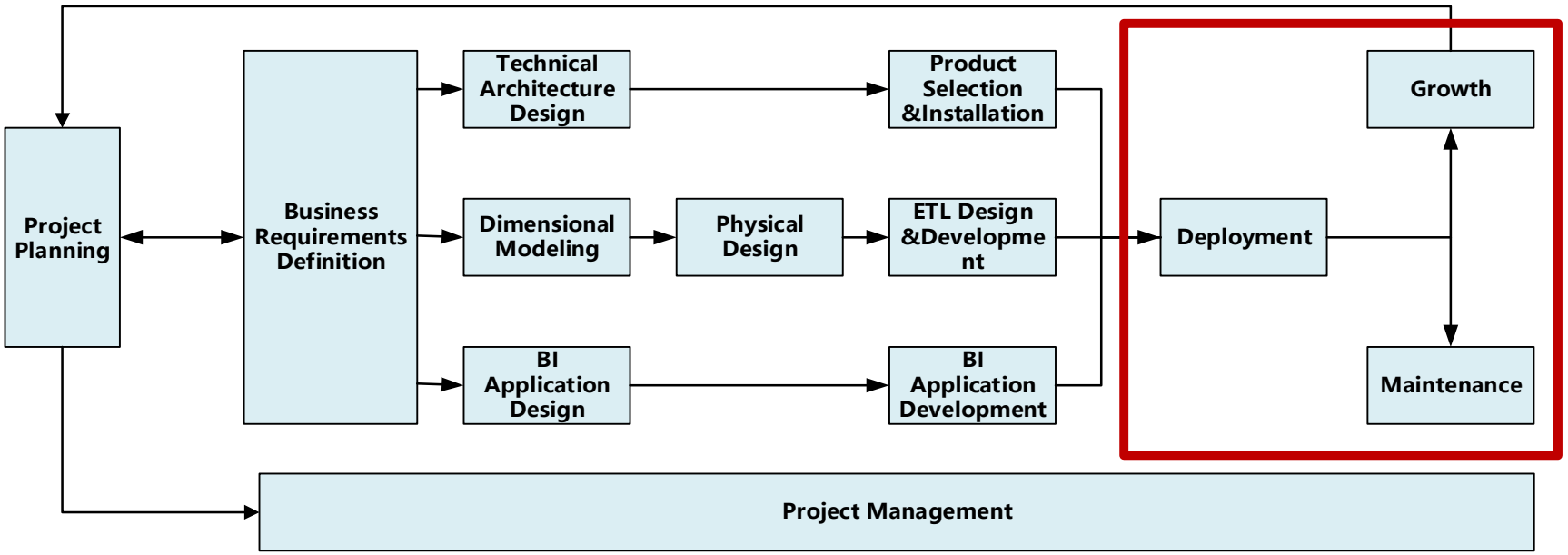
ETL: Extract, Transform, Load

BI application design is based on the needs of business users in data analysis, the design of a series of functional modules, providing query and reporting OLAP（Online Analytical Processing） analysis and data mining tools, so that users can easily access to the required data, and the corresponding processing.

BI应用设计是根据企业用户数据分析方面的需要，设计一系列功能模块，提供查询与报表OLAP（联机分析处理）分析和数据挖掘等工具，使用户能够方便地访问到所需的数据，并进行相应的处理

Resource:《商务智能理论与实践》P33

BI Implementation Process



Project Deploy and Maintenance Phase

ETL: Extract, Transform, Load

The BI system will be submitted to the user to use, at the same time to provide the necessary technical support and user training, maintenance of the BI system to ensure the normal operation of the system, and for the future system upgrade preparation.

将BI系统提交用户使用，同时要提供必要的技术支持与用户培训,对 BI系统进行维护，保障系统的正常运行，并为未来系统升级准备。

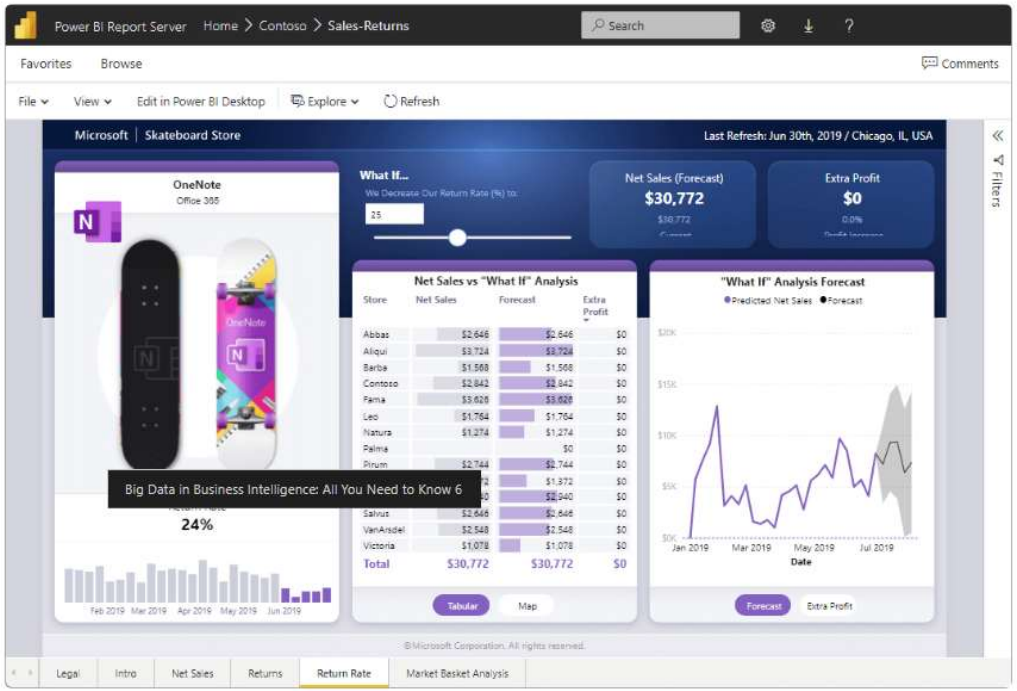
Self-Service BI

Self-service BI and visualization tools are designed to make it easy for non-technical users to analyze and visualize data. These tools allow you to create interactive dashboards and reports, and provide a range of features for exploring and analyzing data.

Some of the most popular self-service BI and visualization tools include Tableau and Power BI. These tools allow you to create stunning visualizations and reports, and provide a range of features for exploring and analyzing data.

Application Case 1:
Retail Industry

应用案例1：零售行业



Resource:https://datarundown.com/bi-big-data/

BI Applications

The application field of BI is very wide, and it has been successfully applied in 13 major industries, such as finance, telecommunication, retail, manufacturing, steel, electronics, health, beer, pharmaceutical, insurance, securities, etc. Typical experience and application cases have been achieved.

BI的应用领域是非常广泛的，已经在金融、电信、零售、制造、钢铁、电子、卫生、啤酒、制药保险、证券等13 大行业取得了成功应用的典型经验和应用案例。

Application Case 2: Retail Industry

应用案例2：零售行业



Resource:https://www.finereport.com/en/dashboard-tools/kpi-dashboard.html

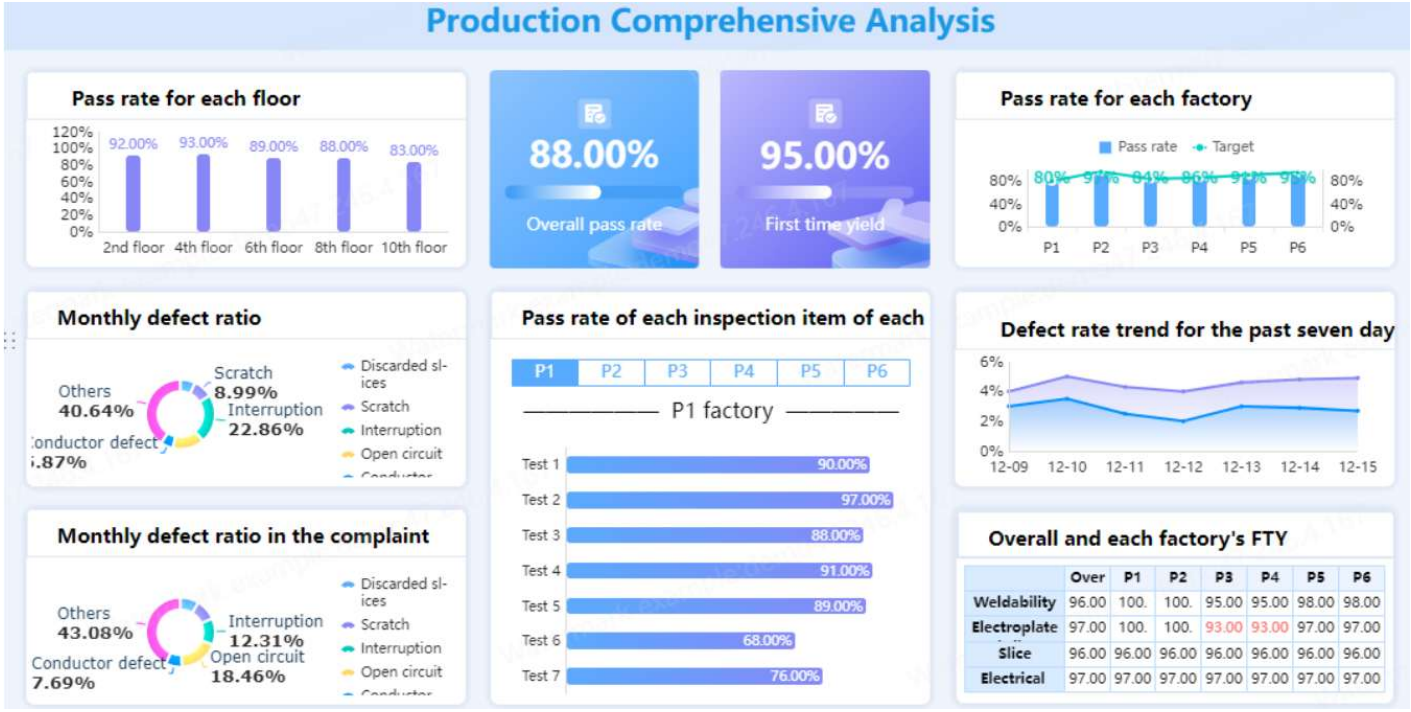
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Application Case 3: Manufacturing Industry

应用案例3：制造行业



Resource:https://www.finereport.com/en/dashboard-tools/kpi-dashboard.html

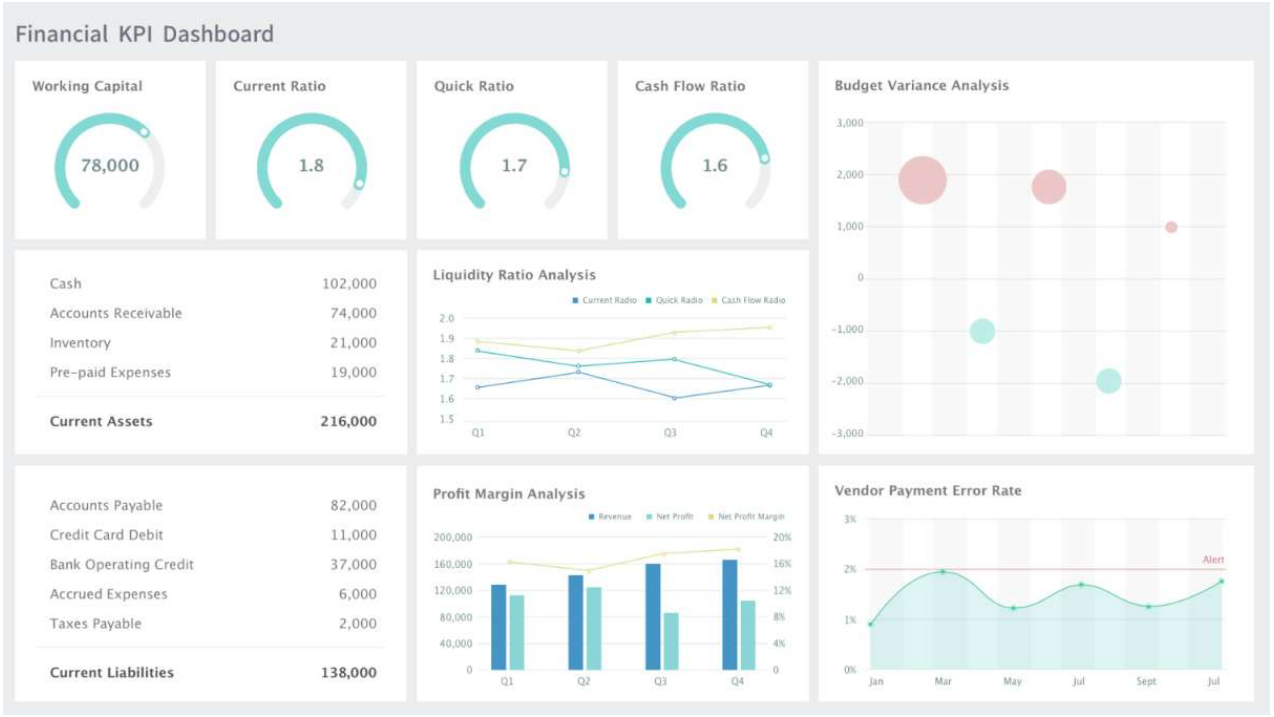
BI Applications

The application field of BI is very wide, and it has been successfully applied in 13 major industries, such as finance, telecommunication, retail, manufacturing, steel, electronics, health, beer, pharmaceutical, insurance, securities, etc. Typical experience and application cases have been achieved.

BI的应用领域是非常广泛的，已经在金融、电信、零售、制造、钢铁、电子、卫生、啤酒、制药保险、证券等13 大行业取得了成功应用的典型经验和应用案例。

Application Case 3: Financial Industry

应用案例3：金融行业



Resource:https://www.finereport.com/en/dashboard-tools/kpi-dashboard.html



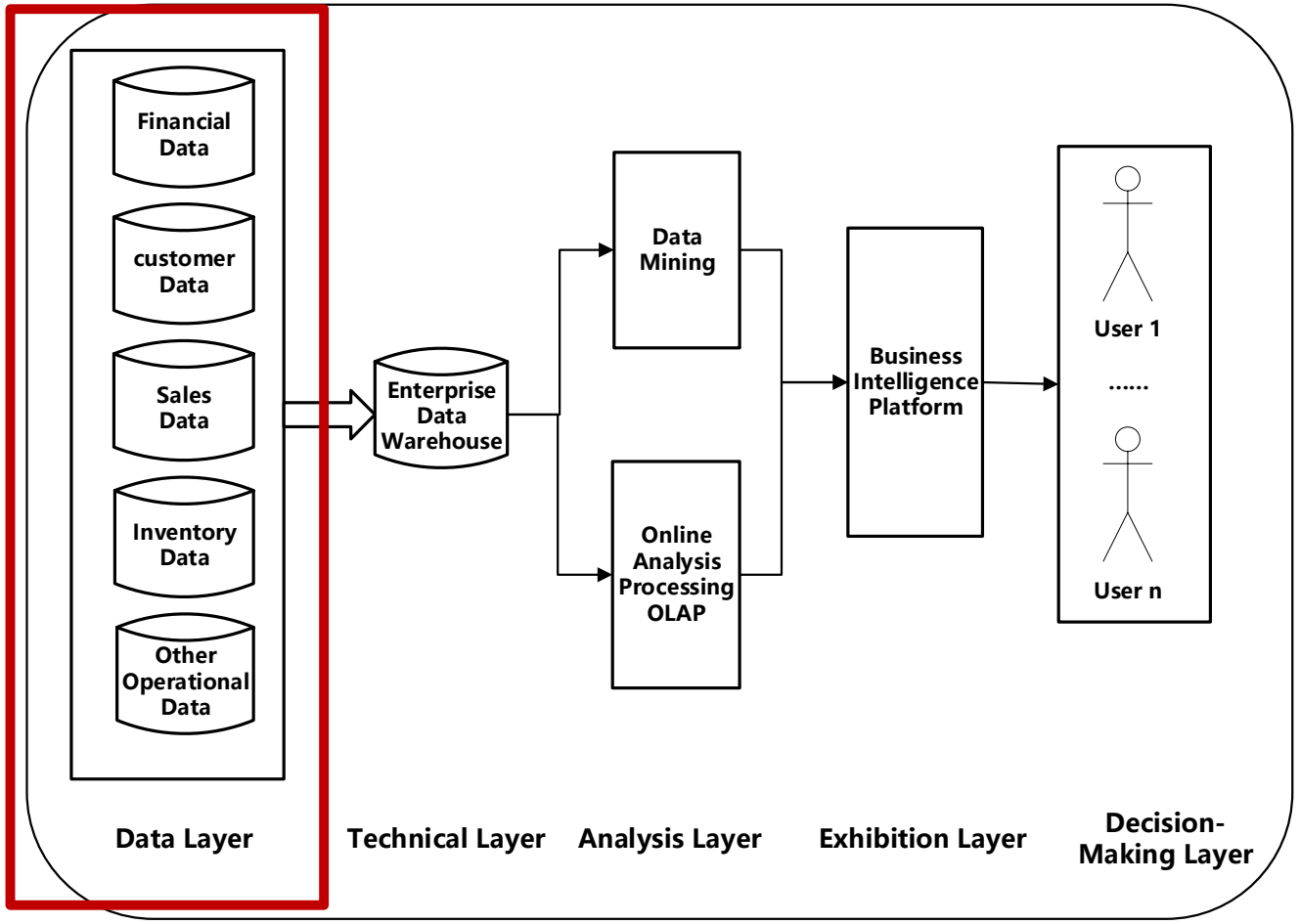
Concepts of layering

Enterprise BI Model

Data Layer:

The data layer ensures that real and effective data is collected from the enterprise's ERP, CRM, Supply Chain Management System SCM and other systems.

数据层:确保从企业的ERP、CRM、供应链管理系统SCM等系统收集到真实有效的数据。

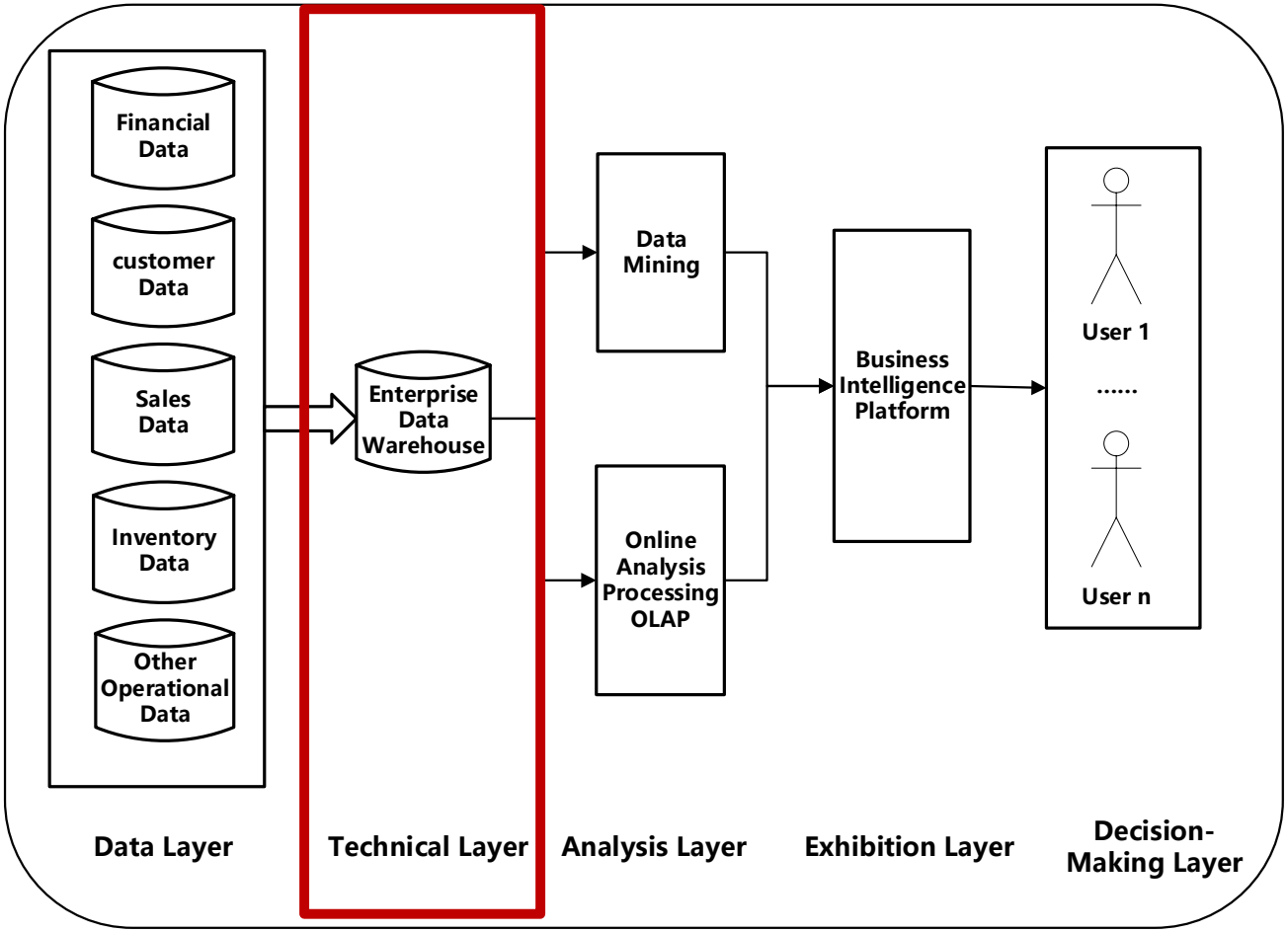


Enterprise BI Model

Technical Layer:

The BI system integrates raw data from the data layer into the data warehouse through ETL, and integrates heterogeneous data for different departments pending further analysis and processing.

技术层：BI系统通过ETL将数据层的原始数据集成到数据仓库中，针对不同部门的异构数据进行整合，以待进一步的分析处理。

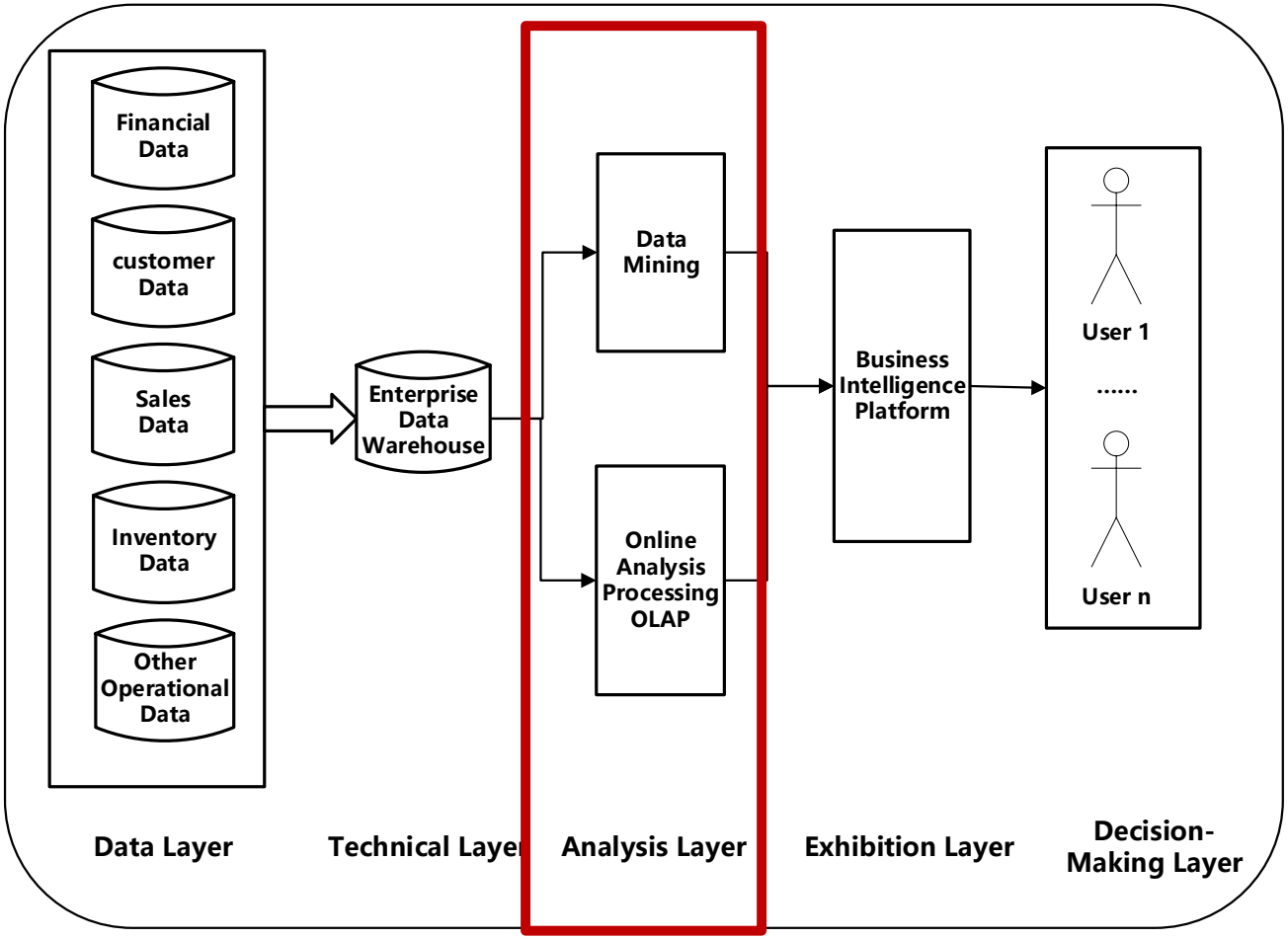


Enterprise BI Model

Analysis Layer:

The system needs to establish a good model library, knowledge base, method library, analyze and mine valuable information from the data warehouse, and transform it into knowledge that users can understand

分析层:系统需建立良好的模型库、知识库、方法库，从数据仓库中分析和挖掘出有价值的信息，转化为用户能理解的知识。

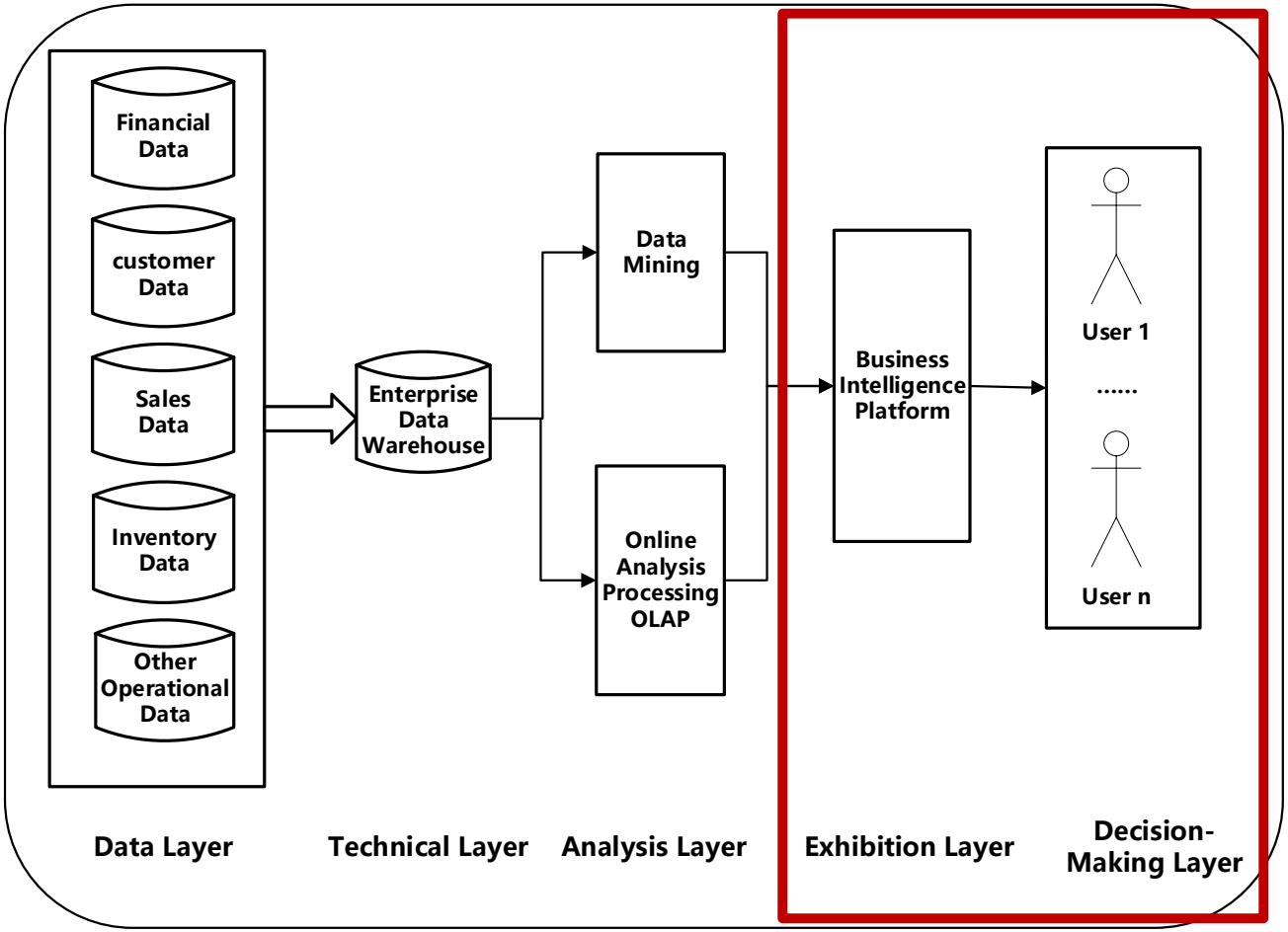


Dashboard in the Exhibition layer

Exhibition Layer:

Enterprises can query reports （**dashboard)**, data graphs, etc., for performance management and other tasks.

展示层：企业可通过查询报表、数据图等，进行绩效管理等工作。



Enterprise BI Model

Exhibition Layer:

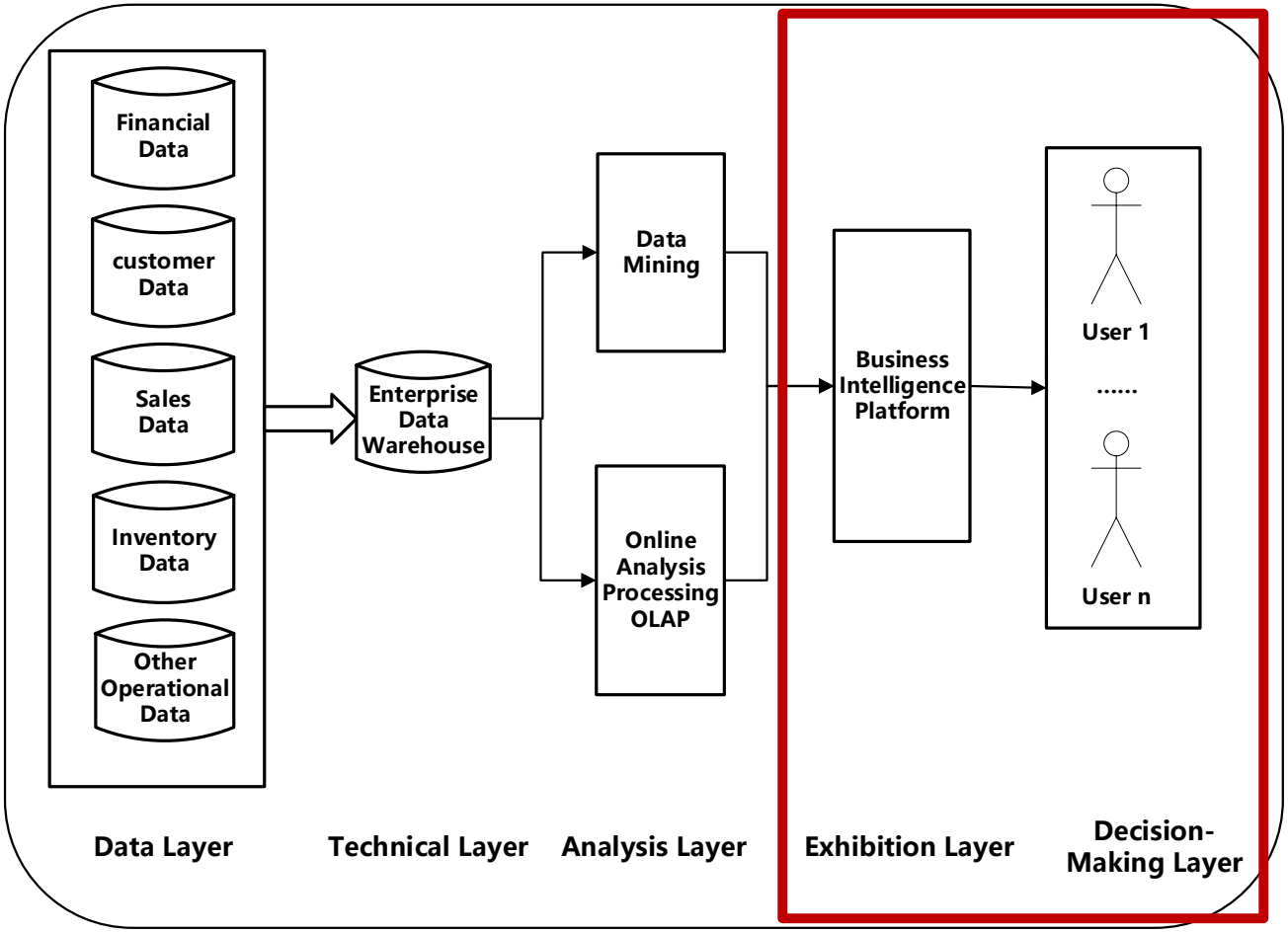
Enterprises can query reports, data graphs, etc., for performance management and other tasks.

展示层：企业可通过查询报表、数据图等，进行绩效管理等工作。

Decision-Making Layer:

Users use the system to provide analytical results, and use strategic decisions to guide specific actions, reflecting the value of BI.

决策层:用户运用分析结果，将战略决策用于指导具体的行动，体现BI的价值。





Thank you!

Contact:
Dr. Hareesh (何玄)

- Final project

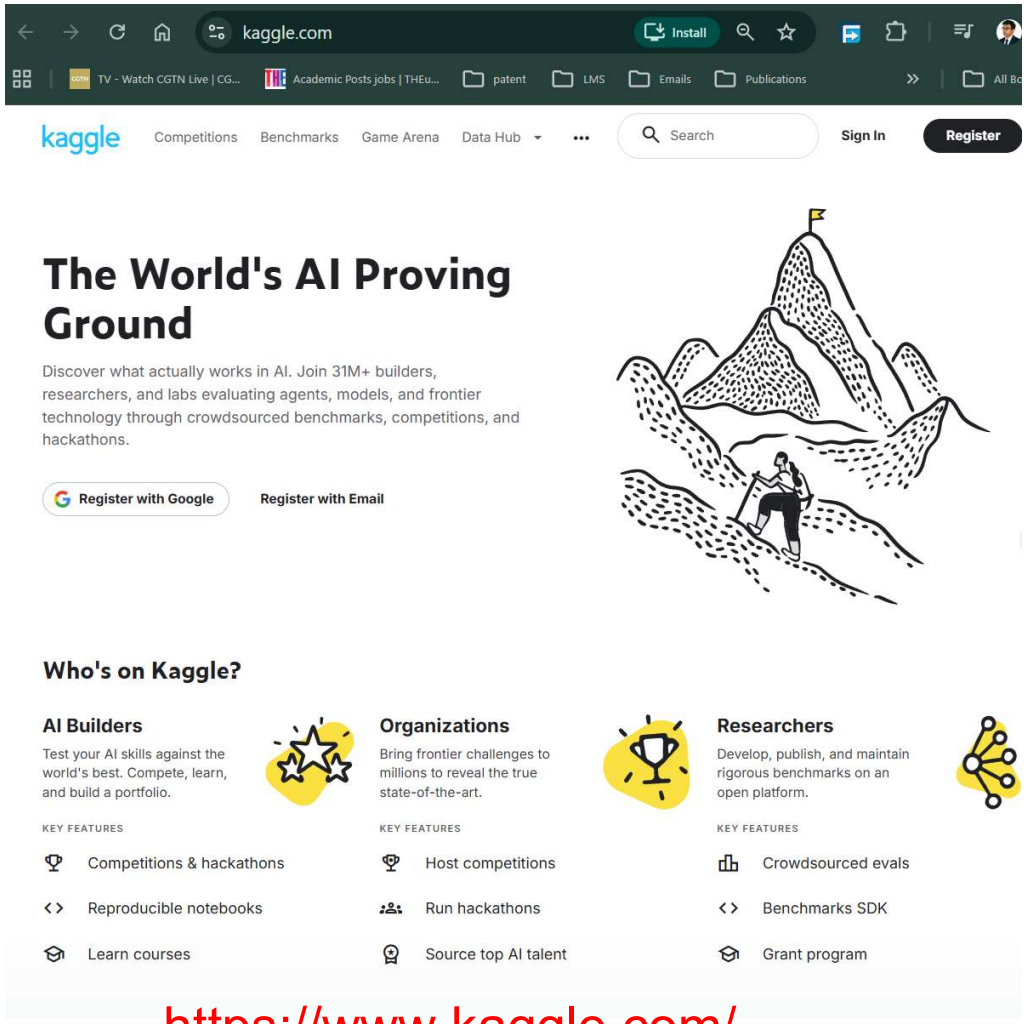
Step 1: Select a Dataset

Each group selects a public dataset from:

Kaggle or Others

Possible domains:

- coffee retail
- food delivery
- logistics
- healthcare
- e-commerce
- banking
- smart cities
- manufacturing



https://www.kaggle.com/